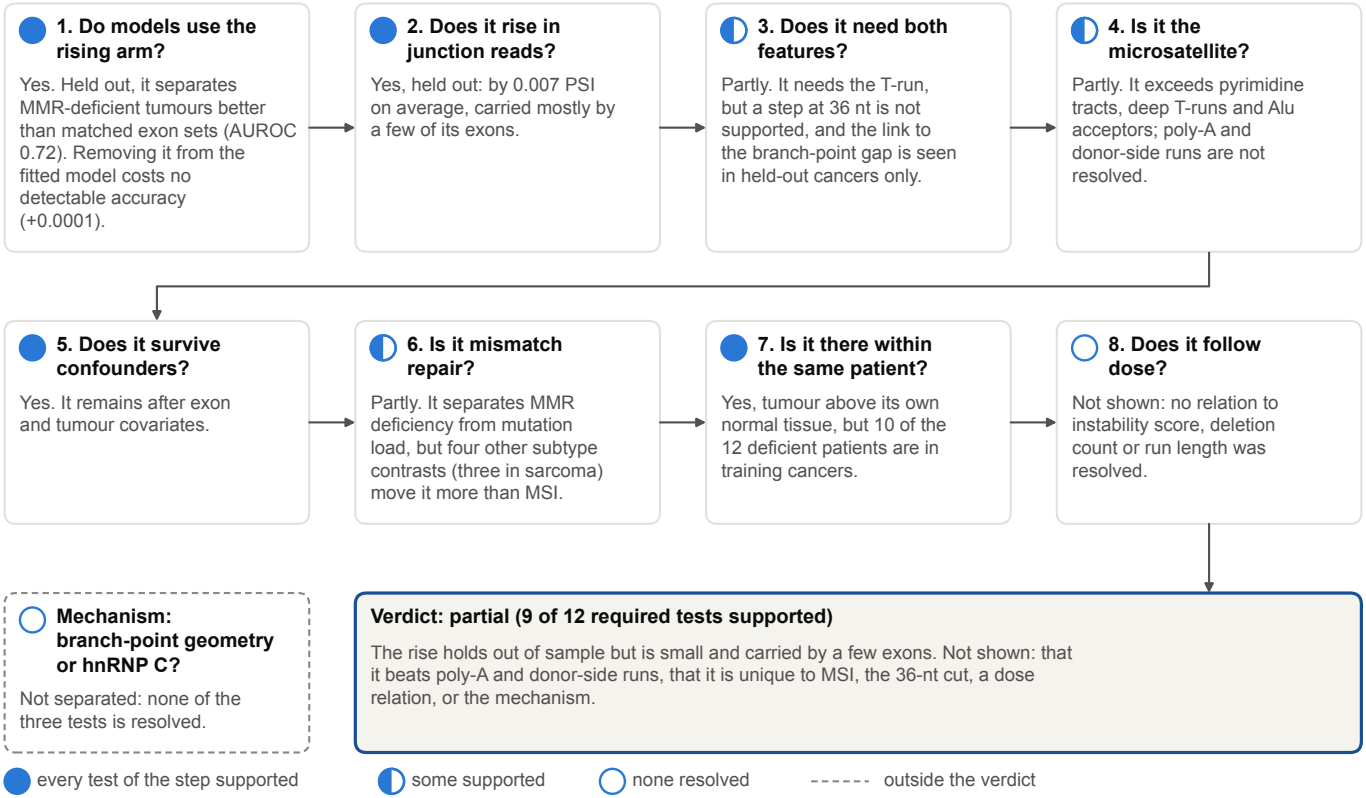
1. Do models use the rising arm?
Yes. Held out, it separates MMR-deficient tumours better than matched exon sets (AUROC 0.72). Removing it from the fitted model costs no detectable accuracy (+0.0001).
2. Does it rise in junction reads?
Yes, held out: by 0.007 PSI on average, carried mostly by a few of its exons.
3. Does it need both features?
Partly. It needs the T-run, but a step at 36 nt is not supported, and the link to the branch-point gap is seen in held-out cancers only.
4. Is it the microsatellite?
Partly. It exceeds pyrimidine tracts, deep T-runs and Alu acceptors; poly-A and donor-side runs are not resolved.
5. Does it survive confounders?
Yes. It remains after exon and tumour covariates.
6. Is it mismatch repair?
Partly. It separates MMR deficiency from mutation load, but four other subtype contrasts (three in sarcoma) move it more than MSI.
7. Is it there within the same patient?
Yes, tumour above its own normal tissue, but 10 of the 12 deficient patients are in training cancers.
8. Does it follow dose?
Not shown: no relation to instability score, deletion count or run length was resolved.
Mechanism: branch-point geometry or hnRNP C?
Not separated: none of the three tests is resolved.
Verdict: partial (9 of 12 required tests supported)
The rise holds out of sample but is small and carried by a few exons. Not shown: that it beats poly-A and donor-side runs, that it is unique to MSI, the 36-nt cut, a dose relation, or the mechanism.
every test of the step supported
some supported
none resolved
outside the verdict